

Foundation (Jalapeno)

- Q1.** If $2^x = 8$, what is the value of x ?
(A) 2
(B) 3
(C) 4
(D) 5
(E) 6
- Q2.** A space lab is experiencing exponential growth of bacteria. If the number of bacteria triples every 24 hours and starts with 100, how many will there be after 3 days? (Assume $y = 100 \cdot 3^x$ where x is in days)
(A) 100
(B) 300
(C) 900
(D) 2700
(E) 8100
- Q3.** What is the value of x if $5^x = 125$?
(A) 2
(B) 3
(C) 4
(D) 5
(E) 6
- Q4.** The radius of a black hole is $10 \cdot 2^x$ meters. If the radius is 40 meters, what is x ?
(A) 1
(B) 2
(C) 3
(D) 4
(E) 5
- Q5.** A physics experiment involves a quantity that decreases exponentially as $y = 200 \cdot (1/2)^x$. If $y = 50$, what is the value of x ?
(A) 1
(B) 2
(C) 3
(D) 4
(E) 5
- Q6.** If $3^x = 81$, what is x ?
(A) 2
(B) 3
(C) 4
(D) 5
(E) 6
- Q7.** A planet's atmosphere is losing pressure exponentially according to $P = 1000 \cdot (1/5)^t$ where t is time in hours. What is the pressure after 2 hours?
(A) 100
(B) 160
(C) 200
(D) 320
(E) 400
- Q8.** Solve for x in the equation $2^{x+1} = 32$.
(A) 3
(B) 4
(C) 5
(D) 6
(E) 7

Open Ended Questions (FRQ)

- Q9.** A comet is traveling through space with an exponential increase in velocity given by $v = 5 \cdot 2^t$, where t is in hours. Find the velocity after 4 hours and show your work.
- Q10.** A chemical reaction in a lab involves a substance decomposing exponentially according to $y = 500 \cdot (1/4)^x$. If after 3 hours, 50 grams remain, find the value of x and explain your reasoning.

Chef's Tips

- Read each question carefully before answering
- Use the space provided for free response questions
- Show all work and explain reasoning where applicable